Broadband raises small-firm productivity by 3.4 percent

Evidence from a staggered municipal broadband rollout

Astrid Lindqvist Tomas Berg

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Demonstration manuscript; setting and data are simulated.

Can universal broadband raise small-firm productivity?

- Small firms dominate employment but adopt technology late (Syverson '11)
- Universal broadband could compress productivity gaps at modest fiscal cost

$\Rightarrow$ **Does broadband causally raise small-firm productivity, and for whom?**

Prior evidence is aggregate or worker-level: (Czernich '13; Akerman '15; Hjort–Poulsen '19).

This paper

We exploit the staggered municipal rollout of Nordavia's National Broadband Program: 284 municipalities, 9,400 small firms, 2008–2019.

1. Broadband raises small-firm labor productivity by **3.4%** (s.e. 0.2 pp)
2. Gains rise with communication intensity, as the model predicts (0.032)
3. The program repays its build cost within six years, small firms alone

Implication: among the cheapest productivity policies available to a small open economy.

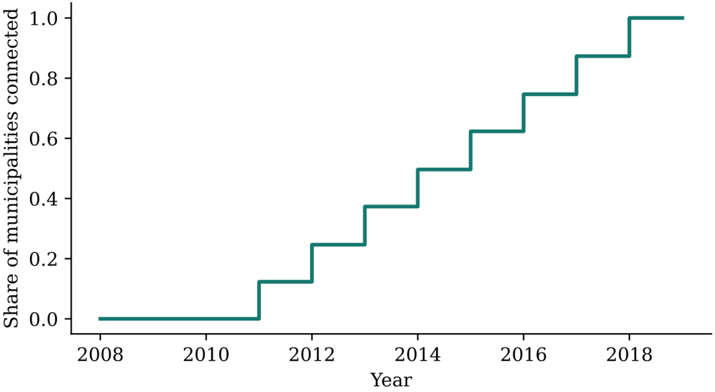
Data

- **Program:** Nordavian National Broadband Program, staggered 2011–2018
- **Sample:** 9,400 firms with 3–49 employees, 2008–2019 (112,800 firm-years)
- **Outcome:** log real value added per worker
- **Comm. intensity:** 2008 wage-bill share of communication jobs (mean 0.42)

Summary stats

Staggered rollout timing identifies the effect

Identifying assumption: arrival timing is unrelated to firms' counterfactual trends

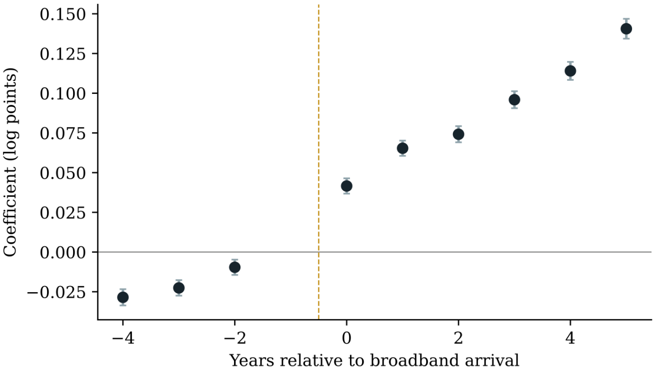


Timing was locked by the 2010 Act, not by the scoring rule's tilt toward high-growth places.

▸ Placebo

Effects appear only after arrival and keep building

Event-study coefficients vs. the year before arrival; 95% CIs, firm-clustered SEs



No jump before arrival; the effect emerges at $t = 0$ and builds. Average gain 3.4% (0.034, s.e. 0.002).

▸ Full table ▸ Robustness

Gains concentrate where the model predicts

Broadband interacted with 2008 communication intensity θ_i

Model prediction: firms doing more communication-intensive work gain more from broadband (Prop. 1: $G'(\theta_i) > 0$).

$$y_{it} - \ell_{it} = \alpha_i + \lambda_t + \beta\, B_{mt} + \underbrace{\gamma\,(B_{mt} \times \theta_i)}_{\text{complementarity, } \gamma>0} + \varepsilon_{it}$$

where $B_{mt} = 1$ once broadband reaches firm i's municipality; β is the average gain, γ the extra gain per unit of communication intensity.

Model confirmed

$\hat{\gamma} = 0.032$ (s.e. 0.005): a firm one SD above mean communication intensity gains about one-fifth more than the average firm.

▸ Model ▸ By intensity

Conclusion

Staggered municipal broadband, 284 municipalities and 9,400 small firms, 2008–2019:

1. Broadband raises small-firm labor productivity by **3.4%** (s.e. 0.2 pp)
2. Gains rise with communication intensity, as the model predicts (0.032)
3. The program repays its build cost within six years, small firms alone

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▸ Policy math

Appendix

Summary statistics

	Mean	SD
Log labor productivity	0.10	0.46
Employees	23.40	7.31
Firm age	21.45	9.56
Communication intensity	0.42	0.20
ICT investment (log)	0.37	0.36

112,800 firm-year observations; 9,400 firms; 2008–2019. Min/max omitted.

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Baseline estimates

Dependent variable: log labor productivity; firm and year fixed effects

	(1) Baseline	(2) +Controls	(3) +Region trends	(4) Muni. cluster
Broadband	**0.034**	0.033	0.030	0.034
	(0.002)	(0.002)	(0.002)	(0.003)
ICT investment (log)		0.000		
		(0.002)		
Firm & year FE	Yes	Yes	Yes	Yes
Observations	112,800	112,800	112,800	112,800
Stable across controls, region trends, and clustering.				

SEs clustered by firm, except col. (4) by municipality. All coefficients significant at 1%.

The headline 3.4% is unchanged by controls and region trends.

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Robustness

Broadband coefficient across specifications; firm and year fixed effects

	(1) Region trends	(2) No controls	(3) Drop 2011–12 cohorts
Broadband	0.030	0.034	0.037
	(0.002)	(0.002)	(0.002)

SEs clustered by firm. Full control set throughout. All significant at 1%.

The effect stays in 0.030–0.037 across reasonable perturbations.

▸ By cohort ◂ Back

Placebo test with permuted rollout years

Rollout years randomly permuted across municipalities; pre-treatment sample

	Pre-period placebo
Placebo broadband (permuted years)	-0.004
	(0.002)

SEs clustered by firm.

A precisely estimated zero: no effect where none should appear.

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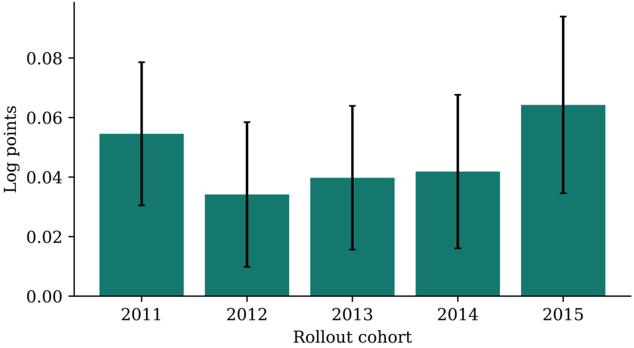
Effects are similar across rollout cohorts

Separate DiD estimate per cohort vs. municipalities connected $\geq$3 years later



Positive and similar across cohorts, so negative weighting is not a concern (Goodman-Bacon '21).

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Heterogeneity by communication intensity

Dependent variable: log labor productivity; firm and year fixed effects

	(1)	(2)
Broadband	0.020	0.034
	(0.003)	(0.002)
Broadband $\times$ comm. intensity	**0.032**	
	(0.005)	
Firm & year FE	Yes	Yes
Observations	112,800	112,800

Communication intensity is the 2008 wage-bill share. SEs clustered by firm. All significant at 1%.

The interaction (0.032) confirms the model's complementarity prediction.

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The model

Task outsourcing: broadband lowers the cost of communication-intensive tasks

- Production: $Y_i = A_i L_i^{1-\eta} \left[(1 - s_i) + \phi\, s_i\right]^{\eta}$, with $\phi > 1$
- Firm outsources a share $s_i \in [0, \theta_i]$ of communication tasks at cost κp
- Sourcing rule: $s_i^* = \theta_i$ if $\phi w \geq \kappa p$, else 0
- Broadband lowers the communication cost $\kappa_0 \to \kappa_1 < \kappa_0$

Proposition 1

The productivity gain is strictly increasing in communication intensity: $G'(\theta_i) > 0$.

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Policy arithmetic

- Aggregate small-firm gain: about 410 million euros per year (valued at 2019 value added)
- Program construction cost: 2.3 billion euros (RDDF cumulative)

⇒ Payback within six years through the small-firm channel alone

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Related literature

Deployed only if asked

- Czernich et al. (2013): broadband and aggregate growth; not the firm margin
- Akerman et al. (2015): broadband is skill-complementary; worker-level, larger firms
- Hjort and Poulsen (2019): fast internet and employment in Africa

This paper: the small-firm segment in an advanced economy, tied to an explicit task-outsourcing model.